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# Jones fractures in elite athletes—2025 international foot and ankle sports consensus and systematic review

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**Abstract**

**Purpose:** Jones fractures are complex injuries with highly variable treatment paradigms. This study aims to establish an expert consensus with support from a systematic review to guide management of Jones fractures in elite athletes.

**Methods:** A consensus process was conducted using a modified Delphi technique with two rounds of questionnaires. Consensus was defined as 75%–85% agreement, strong consensus as 86%–99% agreement, and unanimous consensus as 100% agreement. In addition, a systematic review was conducted to evaluate nonoperative versus operative management of Jones fractures in athletes.

**Results:** Of 37 total consensus statements, 15 achieved unanimous consensus, 21 achieved strong consensus, and 1 reached consensus. The systematic review included 27 studies consisting of 1056 patients. The majority of fractures with a stress aetiology were treated with open reduction internal fixation (86.4%), with a mean return to sport time of 13.3 ± 4.5 weeks and a complication rate of 8.4%. Treatment failure was highest for stress fractures that were treated nonoperatively (8.2%). Traumatic Jones fractures were treated surgically in 62.7% of cases, with the remaining 37.3% managed conservatively. All patients with traumatic fractures successfully returned to sport at an average of 11.6 ± 2.2 weeks regardless of treatment modality. Treatment failure following traumatic fractures was highest for those treated conservatively (16.1%).

**Conclusions:** The average time to return to sport following Jones fracture was shorter among athletes who were treated surgically compared to those who were treated nonoperatively, regardless of injury aetiology or degree of chronicity. This consensus process reached unanimous agreement with respect to the use of MRI for nondisplaced fractures, the use of CT for comminuted or displaced fractures, indications for minimally invasive techniques, and use of adjunctive bone grafting.

**Level of Evidence:** Level V.

**Abbreviations:** BMAC, bone marrow aspirate concentrate; IFASC, International foot and ankle sports consensus; IM, intramedullary; LOE, level of evidence; MINORS, Methodological Index for Non-Randomised Studies; mo, months; M/F, male/female; N, number; nr, not reported; NWB, nonweightbearing; N/A, not applicable; n/r, not recorded; ORIF, open reduction internal fixation; PEDro Scale, physiotherapy evidence database scale; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; ROBINS-I, Risk of bias in non-randomised studies-of interventions; RT, return to; RTS, return to sport; WB, weightbearing.

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**KEYWORDS**
Jones fracture, elite athlete, management, treatment

## INTRODUCTION

Jones fractures in elite athletes can be a challenging orthopaedic injury to treat. These injuries occur at the proximal metaphyseal-diaphyseal junction of the fifth metatarsal, a region where blood supply is poor and bony union is often difficult to achieve [38]. Poor vascularity within this watershed zone is often responsible for delayed healing and prolonged recovery [5]. It is relatively well-accepted that Lawrence type one fractures are managed non-operatively, whereas Lawrence types two and three are managed with early surgical intervention in the athletic population. The present discussion pertains to the diagnosis and management of acute and chronic incomplete Lawrence type 2 fractures, also known as Jones fractures, in elite athletes.

Jones stress fractures are chronic injuries frequently associated with repetitive activity and overuse. Conversely, acute Jones fractures are traumatic injuries often precipitated by a sudden adduction force to the foot. Athletes presenting with an acute fracture typically recall an inciting event. In contrast, patients with a stress fracture may describe insidious pain over the lateral aspect of the foot that has progressively worsened with increased training or activity [20]. A stress aetiology can be confirmed given findings indicative of chronicity on MRI, such as bone marrow oedema or periosteal thickening [15]. Acute fractures are often evidenced by a clear fracture line on plain radiographs that presents without intramedullary sclerosis or significant cortical hypertrophy [38].

The management of Jones fractures in elite athletes remains controversial, and surgery is often claimed as a way to achieve higher rates of radiographic union [1]. However, experts often fail to evaluate treatment outcomes as they relate to athletic performance. Time to return to sport, rate of refracture, and likelihood of treatment success are all important considerations when selecting the most appropriate course of management for an athlete with a Jones fracture.

This study utilises both a systematic review and consensus group to evaluate the different approaches to treatment of Jones fractures in elite athletes. While the systematic review and consensus process were conducted independent of one another, findings from both sources contributed to the development of evidence-based treatment recommendations. Outcomes of the consensus process are reported such that each statement is supplemented with findings from the systematic review where applicable. By bringing together experts from across the globe, this initiative will construct a consensus to help guide clinicians in their management of Jones fractures in elite athletes.

## MATERIALS AND METHODS

### Consensus working group

Thirty-seven global orthopaedic foot and ankle experts were invited to participate and aid in the development of consensus statements for the management of Jones fractures in elite athletes. The participants were invited to be founding members of the International Foot and Ankle Sports Consensus. Surgeons from 25 countries and six continents were selected based on a point system developed by the American Orthopaedic Foot and Ankle Society to evaluate expert credentials. A minimum total score of eight points was required for individuals to be eligible for participation. Scores were assigned by three independent physicians based on the following criteria: (1) academic position, (2) number of academic publications, (3) number of national or international presentations, (4) professional sports team coverage and (5) leadership in a professional foot and ankle or sports medicine society (Table 1).

Expert selection was organised such that a minimum of six continents had representation. Continents with a larger proportion of research output and professional sport involvement were allotted additional positions. Once the expert panel was created, experts were invited to recommend additional individuals for evaluation during a second round of screening based on the aforementioned criteria.

### Consensus survey

Questions were generated by an initial survey and sent to the expert panel. The Delphi method was used to generate consensus statements, and experts completed four rounds of electronic questionnaires. Participant identities were recorded alongside their responses. Questions with significant response variation were chosen for a live debate.

### Consensus criteria

For each survey question, the degree of agreement was expressed as a percentage rounded to the nearest whole number. Consensus was defined as 75%–85%,

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**TABLE 1** Point system used for selection of the expert panel.

| Criteria evaluated | Points awarded |
|---|---|
| Academic position | • Assistant professor: 1 point<br>• Associate professor: 2 points<br>• Full professor: 3 points |
| Number of academic publications | • 1-10 publications: 1 point<br>• 11–20 publications: 2 points<br>• 21 or more publications: 3 points |
| Number of national/international presentations and lectures | • 1–5 presentations/ lectures: 1 point<br>• 6–10 presentations/ lectures: 2 points<br>• 11 or more presentations/ lectures: 3 points |
| Professional sports team coverage | • No coverage: 0 points<br>• Local/regional coverage: 1 point<br>• National/international coverage: 2 points |
| Leadership role in professional foot and ankle/sports medicine society | • No leadership role: 0 points<br>• Member: 1 point<br>• Officer/board member: 2 points |

strong consensus as 86%–99%, and unanimous consensus was indicated by 100%.

## Systematic review

A literature search was conducted in PubMed, EMBASE and The Cochrane Library databases from 2004 to 2024. The search algorithm used in each of the databases was *(Jones) or (metatarsal) and (stress or fracture)*. Any article containing these terms was initially retrieved for article screening. Subsequent evaluation was performed in accordance with the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines [23]. Included articles were those that evaluated the diagnosis and treatment of Jones fractures in athletes, had a minimum follow-up of 3 months, had full text, and were in English. Exclusion criteria were articles that studied paediatric patients, patients with orthopaedic injuries not involving the 5$^{th}$ metatarsal, articles that failed to evaluate treatment outcomes, systematic reviews, meta-analyses, clinical commentaries, books, expert opinion articles, and abstracts without full-text. Any study published in 2003 or earlier was excluded to prioritise recent evidence that reflects the most current knowledge and clinical practices.

## Data extraction and evaluation

All articles populated by the study search terms were retrieved from their respective databases and uploaded to Rayyan, a public software program designed to assist with abstract and title screening. Duplicates were removed by hand. Once complete, selected articles underwent further full-text screening. Articles that met the inclusion criteria were selected for analysis. All studies were assessed by two independent reviewers. The senior author of this paper was consulted to arbitrate any discrepancies that arose.

## Study definitions

Jones fractures were classified according to the mechanism of aetiology. Jones stress fractures were defined as chronic injuries precipitated by repetitive low-grade stress. In contrast, acute Jones fractures were defined as fractures that occurred abruptly following a traumatic event. Studies in the open reduction internal fixation (ORIF) group specifically stated that an incision was made over the fracture site, whereas the studies in the percutaneous group stated percutaneous placement of the screws. Lastly, studies in the conservative treatment group utilised only conservative techniques for fracture management.

## Methodological bias and quality assessment

Overall, 25 articles included in this study were graded via the Methodological Index for Non-Randomised Studies (MINORS) scale due to their observational nature [37]. The MINORS scale categorises studies into comparative or noncomparative groups, with the comparative studies graded from 0 to 24 points and the noncomparative groups graded from 0 to 16 points, with each item worth 0–2 points. The PEDro Scale [26] was used to assess the quality of evidence presented in the remaining two randomised controlled trials. A score of 0–3 is considered ‘poor’, a score of 4–5 is ‘fair’, a score of 6–8 is ‘good’, and a score between 9 and 10 is considered ‘excellent’.

## Statistical analysis

All other statistical analyses were performed using RStudio software (version 4.2.0). Descriptive statistics were calculated for all continuous and categorical variables. Continuous variables were reported as weighted mean and estimated standard deviation, whereas categorical variables were reported as frequencies with percentages.

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# RESULTS

## Overall Consensus

A total of 39 statements were established regarding the diagnostic evaluation, conservative management, surgical treatment and return to sport considerations for Jones fractures in the elite athletic population. Of these, 17 achieved unanimous consensus, 21 achieved strong consensus, and 1 reached consensus. The consensus statements are shown below.

## Consensus statements

*Workup and diagnosis - Jones Fractures*

1. The most common mechanisms of injury for Jones fractures in athletes include overuse, direct impact, twisting, inversion injuries, trauma, and sprains.
   a. Unanimous consensus
2. For Jones fractures, plain radiographs (X-rays) are the most appropriate first-line imaging study to confirm the diagnosis.
   a. Unanimous consensus
3. When a nondisplaced Jones fracture is suspected in an athlete, it is recommended that an MRI and/or CT is ordered initially to confirm the diagnosis.
   a. Unanimous consensus
4. A bone scan is *not* routinely recommended in athletes presenting with suspected Jones fractures.
   a. Strong consensus
5. The clinical workup of Jones fractures should include evaluating recent training loads or changes in athletic activity levels as significant factors, regardless of intended management.
   a. Unanimous consensus
6. For athletes with Jones fractures, assessing for systemic factors such as vitamin D deficiency or osteoporosis should be included as part of the workup.
   a. Strong consensus
7. For displaced Jones fractures, obtaining an MRI to evaluate associated soft tissue injuries or bone marrow oedema is *not* routinely recommended.
   a. Unanimous consensus
8. In an athlete with a Jones fracture, assessment of gait and/or mechanical alignment to identify potential contributing factors, such as poor biomechanics or inappropriate footwear, is recommended.
   a. Unanimous consensus
9. For Jones fractures in athletes, a CT scan is sometimes recommended based on the complexity of the fracture or if the fracture is displaced/comminuted.
   a. Unanimous consensus

*Conservative treatment of chronic incomplete Jones fractures*

10. Indications for conservative management may include chronic, incomplete Jones fractures that are not apparent on MRI.
    a. Strong consensus
11. In chronic, incomplete Jones fractures treated conservatively, a walking boot is recommended over a stiff-soled shoe for immobilisation.
    a. Unanimous consensus
12. For a chronic, incomplete Jones fracture treated conservatively, the athlete may remain non-weight bearing for at least 4–6 weeks.
    a. Strong consensus
13. For a chronic, incomplete Jones fracture treated conservatively, return to partial weight bearing is typically recommended at 6 weeks or following radiological evidence of healing.
    a. Strong consensus
14. In the conservative management of chronic, incomplete Jones fractures, custom insoles/inserts are recommended to correct any biomechanical abnormalities contributing to the injury.
    a. Unanimous consensus
15. For athletes with chronic, incomplete Jones fractures, cross-training in low-impact activities such as swimming or cycling early during the rehabilitation phase is appropriate.
    a. Unanimous consensus
16. An athlete undergoing conservative intervention for a chronic, incomplete Jones fracture can expect to return to light training in their sport within 2–4 months.
    a. Strong consensus
17. An athlete undergoing conservative intervention for a chronic, incomplete Jones fracture can expect to return to full sport (i.e., gameplay) within 4–6 months.
    a. Strong consensus
18. In the case of chronic, incomplete Jones fractures, non-weight bearing physical therapy may be prescribed immediately. Weightbearing physical therapy for strengthening may be started between 4 and 6 weeks after initial presentation.
    a. Unanimous consensus
19. Vitamin D and calcium supplementation is recommended to support bone healing in conservatively managed Jones fractures. Typical vitamin D dosing is 2000 IU daily or 50,000 IU weekly but dosing may vary due to the physician's preference and the individualised treatment of the patient.
    a. Strong consensus
20. If treated conservatively, NSAIDs (e.g., ibuprofen) are NOT used to manage pain or inflammation in athletes with chronic incomplete Jones fractures in the immediate treatment period.
    a. Strong consensus

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21. Conservative treatment for chronic, incomplete Jones fractures does not routinely require DVT prophylaxis during the non-weight bearing phase if there are no risk factors. There may be exceptions due to national and/or hospital requirements.
    a. Strong consensus

*Surgical treatment - Jones Fracture*

22. Indicators for surgical treatment of chronic incomplete Jones fractures or trauma-related Jones fractures in athletes include; displaced fractures, nonunion, failure of conservative management, and fracture gap, high level athletes and need for early return to sport.
    a. Strong consensus
23. Minimally invasive techniques, such as percutaneous screw fixation, are preferred in the surgical treatment of Jones fractures. For non-displaced chronic incomplete Jones fractures, intramedullary screw fixation is typically used for the fixation construct.
    a. Unanimous consensus
24. The fixation construct for a displaced Jones fracture typically involves intramedullary screw fixation.
    a. Strong consensus
25. The fixation construct for a nonunion of the Jones fracture may involve: intramedullary screw fixation with bone grafting, open reduction and internal fixation (ORIF) with bone grafting and plate or percutaneous screw fixation.
    a. Strong consensus
26. In athletes with nonunion chronic incomplete fractures or trauma-related Jones fractures that had been treated conservatively initially, bone grafting (e.g., autograft, allograft) may be recommended in conjunction with surgical fixation.
    a. Unanimous consensus
27. Surgical treatment of chronic incomplete/trauma Jones fractures may involve the use of biologics (e.g., CBMA, PRP, etc.).
    a. Strong consensus
28. Specific fluoroscopic views, including AP, lateral, and oblique, are routinely utilised intraoperatively to assess reduction and fixation.
    a. Strong consensus
29. After surgical fixation of chronic incomplete/trauma-related Jones fractures, 2000–5000 IU of Vitamin D is recommended daily or 50,000 IU can be given weekly.
    a. Strong consensus
30. An athlete may remain non-weight bearing for 3–4 weeks following surgical intervention for a Jones fracture.
    a. Strong consensus
31. An athlete may progress to partial weight bearing between 4 and 6 weeks following surgical intervention for a Jones fracture.
    a. Strong consensus
32. At 6 weeks, an athlete may progress to full weight bearing following surgical intervention for a Jones fracture.
    a. Unanimous consensus
33. An athlete may start PT at 2 weeks following surgical intervention for a Jones fracture.
    a. Strong consensus
34. An athlete undergoing surgical intervention for a Jones fracture can expect to return to light training in their sport within 2–3 months.
    a. Unanimous consensus
35. An athlete undergoing surgical intervention for a Jones fracture can expect to return to full sport (i.e., game play) within 3–6 months.
    a. Strong consensus
36. Biomechanic and rehab assessment, custom orthotics, and identification and management of risk factors are appropriate measures when assessing an athlete's return to play.
    a. Strong consensus
37. Surgical treatment of chronic incomplete/traumatic Jones fractures does not routinely include the use of a bone stimulator postoperatively, although this adjunct may be used to expedite recovery, if available.
    a. Consensus

## Systematic review: initial study results

### Study characteristics

A total of 7688 studies were identified during the initial search. After omission of duplicates, 5821 were screened, 156 were assessed for eligibility, and 27 were ultimately included in this review (2 randomised controlled trials, 12 retrospective studies and 13 case series). Across all studies, 1056 patients met the inclusion criteria. The PRISMA diagram in Figure 1 depicts the full literature search and screening process. The results of bias and quality assessment according to MINORS and the PEDro Scale can be found in Tables 2a and 2b, respectively.

### Systematic review: Patient demographics

A total of 1056 patients (1070 feet) were included in this study. Of these, 699 patients sustained a Jones stress fracture [10, 19, 21, 25, 27, 29, 31, 33, 35, 42] and 357 patients sustained an acute Jones fracture [3, 7, 8, 11, 14, 16, 18, 24, 30, 36, 39, 41]. The mean follow-up period was 32.1 months (range: 11.1–68.7) for patients

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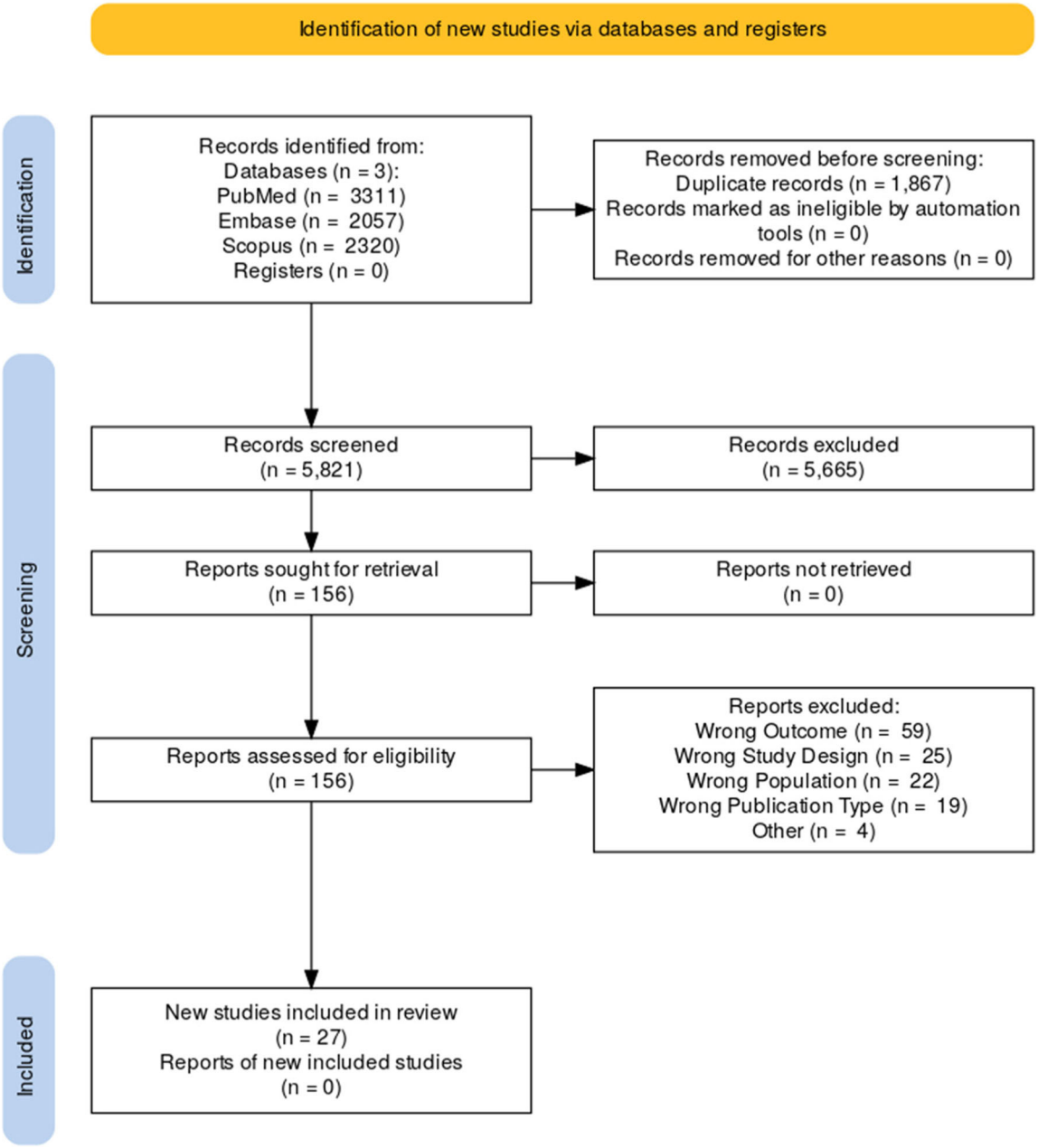


**FIGURE 1** PRISMA 2020 flow diagram outlining the methods utilised in the literature search and screening process.

in the stress fracture group [10, 19, 21, 25, 27, 29, 31, 33, 35, 42] and 25.7 months (range: 3.3–103.3) for patients in the acute fracture group [3, 7, 8, 11, 14, 16, 18, 24, 30, 36, 39, 41].

The average age among patients in the stress fracture group [10, 19, 21, 25, 27, 29, 31, 33, 35, 42] was 21.1 ± 2.6 years, while the average age among patients in the acute fracture group [3, 7, 8, 11, 14, 16, 18, 24, 30, 36, 39, 41] was 29.6 ± 11.1 years. The top three sports played in the stress fracture group [10, 19, 21, 25, 27, 29, 31, 33, 35, 42] were soccer (49.5%), basketball (8.7%) and American football (4.7%). The top three sports played in the acute fracture group [3, 7, 8, 11, 14, 16, 18, 24, 30, 36, 39, 41] were American football (28.3%), soccer (17.4%), and basketball (7.0%). 59.5% of patients in the stress fracture group [10, 19, 21, 25, 27, 29, 31, 33, 35, 42] were high school athletes, 12.7% were recreational athletes, 12.0% were professional athletes, and 10.3% were collegiate athletes. 44.3% of patients in the acute fracture group [3, 7, 8, 11, 14, 16, 18, 24, 30, 36, 39, 41] were recreational athletes, 28.0% were professional athletes, 10.1% were collegiate athletes, and 0.56% were high school athletes. Study characteristics are shown in Table 3.

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**TABLE 2a** Evaluation of study bias and assessment with Methodological Index for Non-Randomised Studies (MINORS) score.

| Study | Clearly stated aim | Inclusion of consecutive patients | Prospective data collection | Endpoints appropriate to study aim | Unbiased assessment of study endpoint | Follow-up sufficiently long | <5% loss to follow-up | Prospective calculation of the study | Adequate control group | Contemporary groups | Baseline efquivalence of groups | Adequate statistical analyses | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| [10] | 2 | 2 | 0 | 2 | 1 | 2 | 2 | 0 | 1 | 2 | 2 | 2 | 18 |
| [21] | 2 | 2 | 1 | 2 | 0 | 2 | 2 | 1 | - | - | - | - | 12 |
| [25] | 2 | 1 | 0 | 2 | 1 | 2 | 0 | 0 | - | - | - | - | 8 |
| [27] | 2 | 2 | 0 | 2 | 1 | 2 | 2 | 0 | - | - | - | - | 11 |
| [28] | 2 | 2 | 0 | 2 | 1 | 2 | 2 | 1 | 2 | 2 | 2 | 2 | 20 |
| [29] | 2 | 2 | 0 | 2 | 1 | 2 | 2 | 0 | - | - | - | - | 11 |
| [31] | 2 | 2 | 1 | 2 | 1 | 2 | 2 | 0 | - | - | - | - | 12 |
| [32] | 2 | 2 | 0 | 2 | 1 | 2 | 2 | 0 | 2 | 1 | 2 | 2 | 18 |
| [35] | 2 | 2 | 0 | 2 | 1 | 1 | 2 | 0 | - | - | - | - | 10 |
| [42] | 2 | 2 | 0 | 2 | 1 | 2 | 2 | 0 | - | - | - | - | 11 |
| [19] | 2 | 1 | 0 | 2 | 1 | 2 | 0 | 0 | - | - | - | - | 8 |
| [3] | 2 | 2 | 1 | 2 | 0 | 2 | 0 | 0 | - | - | - | - | 9 |
| [11] | 2 | 2 | 0 | 2 | 0 | 2 | 2 | 0 | - | - | - | - | 10 |
| [12] | 2 | 1 | 0 | 2 | 1 | 2 | 2 | 0 | 2 | 2 | 2 | 1 | 17 |
| [16] | 2 | 2 | 0 | 2 | 0 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 18 |
| [39] | 2 | 1 | 0 | 2 | 1 | 2 | 2 | 0 | 1 | 2 | 0 | 1 | 14 |
| [7] | 2 | 2 | 0 | 2 | 1 | 2 | 2 | 0 | - | - | - | - | 11 |
| [8] | 2 | 1 | 0 | 2 | 0 | 2 | 2 | 0 | - | - | - | - | 9 |
| [13] | 2 | 1 | 0 | 2 | 0 | 2 | 2 | 0 | - | - | - | - | 9 |
| [14] | 2 | 2 | 0 | 2 | 1 | 2 | 2 | 0 | - | - | - | - | 11 |
| [18] | 2 | 2 | 0 | 2 | 1 | 2 | 2 | 0 | - | - | - | - | 11 |
| [30] | 2 | 1 | 0 | 2 | 1 | 2 | 2 | 0 | - | - | - | - | 10 |
| [36] | 2 | 0 | 0 | 2 | 0 | 2 | 2 | 0 | - | - | - | - | 8 |
| [40] | 2 | 2 | 0 | 2 | 1 | 1 | 2 | 0 | - | - | - | - | 10 |
| [41] | 2 | 2 | 0 | 2 | 1 | 2 | 0 | 0 | - | - | - | - | 9 |

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**TABLE 2b** Evaluation of study bias and assessment with PEDro scale.

| Study | [24] | [33] |
|---|---|---|
| 1. Specification of eligibility criteria | 1 | 1 |
| 2. Random allocation | 1 | 1 |
| 3. Concealed allocation | 1 | 0 |
| 4. Prognostic similarity at baseline | 1 | 1 |
| 5. Participant blinding | 0 | 0 |
| 6. Therapist blinding | 0 | 0 |
| 7. Assessor blinding | 0 | 0 |
| 8. >85% follow-up of at least one key outcome | 1 | 1 |
| 9. Intention to treat analysis | 0 | 0 |
| 10. Between-group statistical comparison for at least one key outcome | 1 | 1 |
| 11. Point estimates and measures of variability for at least one key outcome | 1 | 1 |
| Total score | 7/10 | 6/10 |

## Systematic review: Surgical versus nonsurgical treatment

Operative and non-operative treatment outcomes were compared for both stress and acute Jones fractures. Approaches to treatment included ORIF, percutaneous fixation, suture anchor fixation, or conservative management (Figure 2). Studies in the ORIF group stated that an open incision was made over the fracture site, whereas studies in the percutaneous fixation group stated percutaneous placement of screws and/or k-wires.

Of those patients treated for a stress Jones fracture, 93.0% underwent surgical intervention [10, 21, 25, 27, 29, 31, 33, 35, 42] and 7.0% were managed conservatively [10, 19, 33]. Treatment consisted of ORIF [10, 21, 27, 29, 31, 32, 35, 42] in 604 patients (86.4%), percutaneous fixation [25, 33] in 46 patients (6.6%), and conservative treatment [10, 19, 33] in 49 patients (7.0%). Hardware utilised for operative fixation included screws [21, 25, 27, 29, 31, 33, 35] in 94.2% of patients and plantar plates [42] in 5.8% of patients. Among those who received surgical intervention, use of bone marrow aspirate concentrate (BMAC) [21, 27, 28] was reported in 216 patients (33.2%), and placement of a bone graft [21, 27, 28, 35, 42] was reported in 419 patients (64.5%). Conservative treatment consisted of short-leg cast or boot immobilisation in 40 patients (81.6%) [10, 19] and electroshockwave therapy [33] followed by brace immobilisation in 9 patients (18.3%).

Of those treated for acute Jones fractures, 62.7% were treated surgically [7, 11, 14, 16, 18, 24, 30, 36, 39, 41] and 37.3% were managed conservatively [3, 8, 12, 16, 24]. Treatment involved ORIF [7, 12, 14, 36, 39] in 93 patients (26.1%), percutaneous fixation [11, 16, 18, 24, 30, 40, 41] in 127 patients (35.6%), suture anchor fixation [13] in four patients (1.1%), and conservative management [3, 8, 12, 16, 24] in 133 patients (37.3%). Hardware utilised for operative fixation consisted of screws [3, 8, 11, 14, 16, 18, 24, 30, 39, 41] in 95.0% of patients, plantar plates [7] in 3.6% of patients, and a locking plate [36] in 1.4% of patients. A 5.5 mm suture anchor was used in all four patients who underwent suture anchor fixation [13]. Use of BMAC [14, 18, 30] was reported in 18.2% of surgical cases (40 patients), and placement of a bone graft [7, 14, 18, 30, 36] was reported in 22.7% of cases (50 patients). Conservative treatment consisted of cast immobilisation [16, 24] in 48.1% of patients (64 out of 133), a walking boot [3, 8] in 34.6% of patients (46 out of 133), and shock wave therapy [12] in 17.3% of patients (23 out of 133). Platelet-rich plasma was used in 5.3% of patients who were treated conservatively (7 out of 133) [8].

## Systematic review: Treatment outcomes

All patients with Jones stress fractures [10, 19, 21, 25, 27, 29, 31, 33, 35, 42] returned to sport regardless of treatment intervention. The weighted mean time to return to sport was 13.3 ± 4.5 weeks for the ORIF group [10, 21, 27, 29, 31, 32, 35, 42], 8.7 ± 3.6 weeks for the percutaneous fixation group [25, 33], and 18.2 ± 7.7 weeks for the conservative treatment group [10, 19, 33]. Overall, 90.1% of patients in the ORIF group [10, 21, 27, 29, 31, 32, 35, 42] achieved full union (526 out of 584) at a weighted mean time to union of 13.4 ± 9.9 weeks. Furthermore, 100% of patients in the percutaneous fixation group [25, 33] achieved full union (46 out of 46), however, time to union could not be extracted. Within the conservative group [10, 19, 33], 87.8% of patients achieved full union (43 out of 49) at a weighted mean time to union of 21.9 ± 3.4 weeks.

Among patients in the acute Jones fracture group [3, 7, 8, 11, 14, 16, 18, 24, 30, 36, 39, 41], return to sport was reported in 98.6% of patients in the ORIF group [7, 12, 14, 36, 39] (71 out of 72), 100% of patients in the percutaneous group [11, 16, 18, 24, 30, 40, 41] (78 out of 78), 100% of patients in the suture anchor cohort [13] (4 out of 4), and 100% of patients in the conservative cohort [3, 8, 12, 16, 24] (30 out of 30). The weighted mean time to return to sport was 12.8 ± 2.9 weeks in the ORIF group [7, 12, 14, 36, 39], 8.5 ± 2.9 weeks in the percutaneous group [11, 16, 18, 24, 30, 40, 41], 13 weeks in the suture anchor group [13], and 10.4 ± 3.8 weeks in the conservative treatment group [3, 8, 12, 16, 24]. Furthermore, 94.6% of patients in the ORIF cohort achieved full union (87 out of 92) at a weighted mean time to union of 5.8 ± 1.8 weeks [7, 12, 14, 36, 39]. Within the

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**TABLE 3** Study characteristics and patient demographics.

| Author | LOE | *N* Patients | *N* feet | Follow-up (months) | Age (years) | M/F | Level of play |
|---|---|---|---|---|---|---|---|
| Jones stress fracture group | | | | | | | |
| [33] | 2 | 18 | 18 | 33 | 21.5 | 18/0 | College (*n* = 18) |
| [10] | 4 | 61 | 61 | 40 | 27.3 | 46/15 | Elite (*n* = 17)<br>Recreational (*n* = 35) |
| [21] | 4 | 37 | 37 | 60.6 | 23 | 37/0 | Professional (*n* = 37) |
| [25] | 4 | 37 | 37 | 27.9 | 19.3 | 29/8 | High school (*n* = 37) |
| [27] | 4 | 73 | 78 | 12.7 | 19.3 | 74/4 | Professional (*n* = 6)<br>Recreational (*n* = 72) |
| [28] | 4 | 95 | 101 | nr | 19.5 | 96/5 | High school (*n* = 95) |
| [29] | 4 | 60 | 60 | 40.1 | 19 | 59/1 | Professional (*n* = 20)<br>College (*n* = 25)<br>High school (*n* = 13)<br>Recreational (*n* = 2) |
| [32] | 4 | 20 | 20 | 16.5 | 24.7 | 15/5 | Professional (*n* = 2)<br>College (*n* = 8)<br>High school (*n* = 3)<br>Recreational (*n* = 7) |
| [31] | 4 | 23 | 24 | 19.8 | 20 | 17/6 | Professional (*n* = 2)<br>College (*n* = 8)<br>High school (*n* = 8)<br>Recreational (*n* = 6) |
| [35] | 4 | 222 | 222 | 11.1 | 18.8 | 203/19 | High school (*n* = 222) |
| [42] | 4 | 38 | 38 | 23 | 19.7 | 37/1 | High school (*n* = 38) |
| [19] | 4 | 15 | 15 | 68.7 | 21.2 | 15/0 | College (*n* = 352)<br>Recreational (*n* = 2) |
| Jones acute fracture group | | | | | | | |
| [24] | 1 | 37 | 37 | 25.3 | 25.6 | 35/2 | Recreational (*n* = 37) |
| [3] | 3 | 39 | 39 | 22 | 40 | 18/21 | Recreational (*n* = 39) |
| [11] | 3 | 21 | 21 | 12 | 23 | 21/0 | Professional (*n* = 21) |
| [12] | 3 | 43 | 43 | nr | nr | 21/22 | College (*n* = 1)<br>Recreational (*n* = 1) |
| [16] | 3 | 69 | 69 | 3.3 | 50.5 | 32/37 | Recreational (*n* = 69) |
| [39] | 3 | 40 | 41 | nr | nr | 40/0 | Profesional (*n* = 40) |
| [7] | 4 | 8 | 8 | 38.4 | 21.9 | 8/0 | Professional (*n* = 6)<br>College (*n* = 2) |
| [8] | 4 | 7 | 7 | 6 | 19 | 7/0 | College (*n* = 7) |
| [13] | 4 | 4 | 4 | 27.3 | 45.4 | 1/3 | Recreational (*n* = 4) |
| [14] | 4 | 21 | 21 | 29 | 28.4 | 16/5 | nr |
| [18] | 4 | 25 | 25 | 6 | 24 | 25/0 | Professional (*n* = 25) |
| [30] | 4 | 8 | 8 | 12 | 25.7 | 8/0 | Professional (*n* = 8) |
| [36] | 4 | 3 | 3 | 42 | 18.5 | 1/1 | High School (*n* = 2) |
| [40] | 4 | 7 | 7 | 7.5 | 43.3 | nr | College (*n* = 1)<br>Recreational (*n* = 6) |
| [41] | 4 | 25 | 26 | 103.3 | 20 | 22/3 | College (*n* = 25) |

Abbreviations: LOE, level of evidence; M/F, male/female; *N*, number; nr, not recorded.

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**FIGURE 2** (a) AP weightbearing x-ray of the right foot demonstrating a Jones fracture. (b) Lateral weightbearing X-ray of the right foot demonstrating screw fixation for Jones fracture. (c) Oblique X-ray of the right foot showing screw fixation for Jones fracture.

percutaneous fixation group, 98.7% of patients achieved full union (78 out of 79) at a weighted mean time to union of 8.8 ± 2.0 weeks [11, 16, 18, 24, 30, 40, 41]. Furthermore, 100% of patients in the suture anchor group achieved full union (4 out of 4) at a weighted mean time to union of 13 weeks [13]. Lastly, 86.1% in the conservative cohort achieved full union (81 out of 94) at a weighted mean time to union of 11.2 ± 2.5 weeks [3, 8, 12, 16, 24]. The average length of immobilisation among patients who underwent operative fixation was 2.1 ± 1.4 weeks (range: 0-5) [7, 11, 14, 16, 18, 24, 30, 36, 39, 41]. The average length of immobilisation among patients who underwent conservative treatment was 4.2 ± 3.5 weeks (range: 2-11) [3, 8, 12, 16, 24].

## Systematic review: Complications, failures, and secondary surgical procedures

Within the stress Jones fracture group, complications were reported in 8.4% of patients in the ORIF group [10, 21, 27, 29, 31, 32, 35, 42] (51 out of 604), 5.5% of patients in the percutaneous fixation group [25, 33] (1 out of 18), and 4.1% of patients in the conservative treatment group [10, 19, 33] (2 out of 49). The greatest number of treatment failures occurred in the conservative group, in which 8.2% of patients (4 out of 49) failed treatment [10, 19, 33]. Failure rates in the ORIF [10, 21, 27, 29, 31, 32, 35, 42] and percutaneous group [25, 33] were 6.5% (39 out of 604) and 4.3% (2 out of 46), respectively. Secondary surgical procedures were performed in 3.1% of patients in the ORIF group [10, 21, 27, 29, 31, 32, 35, 42] (19 out of 604), 5.5% of patients in the percutaneous fixation group [25, 33] (1 out of 18), and 12.2% of patients in the conservative group [10, 19, 33] (6 out of 49). The results of this analysis are presented in Table 4a.

Within the acute Jones fracture group, complications occurred in 11.0% of patients in the ORIF group [7, 12, 14, 36, 39] (10 out of 92), 11.8% of patients in the percutaneous fixation group [11, 16, 18, 24, 30, 40, 41] (15 out of 127), 0% of patients in the suture anchor group [13] (0 out of 4), and 10.7% of patients in the conservative group [3, 8, 12, 16, 24] (11 out of 103). The failure rate was highest among those who underwent conservative treatment, with 14 out of 87 patients reporting treatment failure (16.1%) [3, 8, 12, 16, 24]. The rate of failure among patients in the ORIF [7, 12, 14, 36, 39], percutaneous fixation [11, 16, 18, 24, 30, 40, 41], and suture anchor [13] groups was 5.4% (5 out of 92), 3.1% (4 out of 124), and 0.0% (0 out of 4), respectively. Secondary surgical procedures were performed in 14.1% of patients in the ORIF group [7, 12, 14, 36, 39] (13 out of 92), 12.6% of patients in the percutaneous group [11, 16, 18, 24, 30, 40, 41], (16 out of 127), 0.0% of patients in the suture anchor group [13] (0 out of 4), and 3.9% of patients in the conservative group [3, 8, 12, 16, 24] (4 out of 103). The results of this analysis are presented in Table 4b.

# DISCUSSION

The most important findings of this study pertain to the updated recommendations on Jones fracture management in both surgical and conservative capacities. The radiologic work-up of a Jones fracture, as formulated through unanimous consensus, should consist of plain radiographs followed by MRI for nondisplaced

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**TABLE 4a** Complications, failures, and secondary surgical procedures: Jones stress fracture group.

| Author | *N* Patients | *N* Feet | Nonunion (*n*) | Partial union (*n*) | Delayed union (*n*) | Refracture (*n*) | Unable to RTS (*n*) | Failures | Complications | Secondary surgical procedures | Hardware removal |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Open reduction internal fixation (ORIF) | | | | | | | | | | | |
| [27] | 73 | 78 | 2 | 0 | 0 | 1 | 0 | 3 | 1 | 0 | 0 |
| [28] | 95 | 101 | 2 | nr | nr | nr | nr | 2 | 1 | 3 | nr |
| [21] | 37 | 37 | 1 | 0 | 8 | 1 | 0 | 9 | 6 | 1 | 0 |
| [35] | 222 | 222 | 10 | 0 | 2 | 19 | nr | 12 | 22 | nr | 0 |
| [31] | 23 | 24 | 0 | 6 | nr | 0 | 0 | 6 | 0 | 0 | 0 |
| [32] | 20 | 20 | 0 | nr | nr | 3 | 0 | nr | 3 | 0 | 0 |
| [10] | 36 | 36 | 4 | 0 | 0 | nr | nr | 4 | 5 | 5 | 5 |
| [29] | 60 | 60 | 1 | 0 | 1 | 0 | 0 | 2 | 0 | 1 | 1 |
| [42] | 38 | 38 | 0 | 0 | 0 | 4 | 1 | 1 | 13 | 9 | 9 |
| Percutaneous fixation | | | | | | | | | | | |
| [25] | 37 | 37 | 0 | 0 | 0 | nr | 0 | 0 | nr | 0 | nr |
| [33] | 18 | 18 | 0 | 0 | 2 | 1 | 0 | 2 | 1 | 1 | 0 |
| Conservative treatment | | | | | | | | | | | |
| [19] | 15 | 15 | 2 | 0 | 2 | 1 | 3 | 4 | 2 | 2 | n/a |
| [33] | 9 | 9 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | n/a |
| [10] | 25 | 25 | 4 | 0 | 0 | nr | nr | 4 | 0 | 4 | n/a |

*Note*: Failures were defined as nonunion, delayed union, partial union. Complications were defined as refracture, chronic pain, hardware breakage or irritation, postoperative infection, or nerve injury.
Abbreviations: *N*, number; RTS, Return to sport.

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**TABLE 4b** Complications, failures, and secondary surgical procedures: Jones acute fracture group.

| Author | *N* Patients | *N* Feet | Nonunion (*n*) | Partial union (*n*) | Delayed union (*n*) | Refracture (*n*) | Unable to RTS (*n*) | Failures | Complications | Secondary surgical procedures | Hardware removal |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Open reduction internal fixation (ORIF) | | | | | | | | | | | |
| [39] | 40 | 41 | 3 | nr | nr | 1 | 0 | 3 | 2 | 1 | 1 |
| [12] | 20 | 20 | 2 | nr | nr | 1 | nr | 2 | 2 | 9 | 9 |
| [14] | 21 | 21 | 0 | 0 | 0 | 1 | 0 | 0 | 4 | 3 | 3 |
| [7] | 8 | 8 | 0 | nr | 0 | nr | 0 | 0 | 2 | 0 | nr |
| [36] | 2 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | nr |
| Percutaneous fixation | | | | | | | | | | | |
| [30] | 7 | 7 | 0 | nr | nr | 3 | 0 | 0 | 3 | 3 | nr |
| [41] | 25 | 26 | 0 | nr | nr | 1 | 0 | 0 | 3 | 4 | 3 |
| [11] | 21 | 21 | 1 | nr | nr | 1 | 0 | 1 | 1 | 1 | nr |
| [40] | 7 | 7 | 1 | 0 | 0 | nr | nr | 1 | 1 | 1 | 0 |
| [18] | 25 | 25 | 0 | nr | nr | 4 | 0 | 0 | 4 | 3 | 3 |
| [24] | 19 | 19 | 1 | nr | nr | 0 | nr | 1 | 1 | 4 | 3 |
| [16] | 23 | 23 | 1 | nr | 0 | nr | nr | 1 | 2 | nr | nr |
| Suture anchor fixation | | | | | | | | | | | |
| [13] | 4 | 4 | 0 | nr | nr | nr | 0 | 0 | 0 | 0 | 0 |
| Conservative treatment | | | | | | | | | | | |
| [3] | 39 | 39 | 0 | nr | nr | nr | nr | nr | 7 | 0 | 0 |
| [16] | 46 | 46 | 0 | nr | 5 | nr | nr | 5 | 0 | 0 | 0 |
| [8] | 7 | 7 | nr | nr | nr | nr | nr | nr | nr | nr | nr |
| [24] | 18 | 18 | 5 | nr | 1 | 2 | nr | 6 | 4 | 4 | 0 |
| [12] | 23 | 23 | 2 | nr | 1 | nr | nr | 3 | nr | nr | nr |

*Note*: Failures were defined as nonunion, delayed union, partial union. Complications were defined as refracture, chronic pain, hardware breakage or irritation, postoperative infection, or nerve injury.
Abbreviations: *N*, number; RTS, Return to sport.

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fractures or CT scan for comminuted or displaced fractures. It is important to note that MRIs are not necessary to evaluate for soft tissue injury or bone marrow oedema in the setting of displaced fractures. Furthermore, strong consensus was reached that bone scans are not routinely performed for nondisplaced stress fractures.

Beddard et al. [4] demonstrated that footwear is an important factor in both acute and stress Jones fractures. The authors of this 2024 study found that studded footwear can contribute to abnormal loading of the fifth metatarsal if the width of the foot is allowed to overhang the width of the studs. Additionally, biomechanical abnormalities, such as flat feet or leg length discrepancy, have been shown to be associated with an increased risk of Jones stress fractures [6]. All members of this consensus group agreed that a formal assessment of footwear and gait mechanics should be performed to evaluate for potential contributing factors in any athlete who presents with a Jones fracture. Additionally, it was agreed with strong consensus that all athletes with stress fractures should be evaluated for vitamin D deficiency and osteoporosis. The outcomes of this consensus are further supported by the current literature which has shown that consistently low vitamin D levels along with high-intensity training increase an athlete's risk for stress fracture [22].

Acuity is an important factor when considering operative versus nonoperative treatment in an elite athlete with a Jones fracture. Jones stress fractures are at high risk of fracture propagation, displacement, and nonunion [34]. For this reason, surgical intervention is often preferred in athletes with demanding training schedules and high levels of physical activity [4]. Among this consensus group, there was strong agreement that fractures with evidence of displacement, nonunion, or failed conservative treatment should be treated operatively, along with those that occur in high-level athletes with a need for early return to play. Furthermore, the results of this systematic review found that athletes with a stress Jones fracture were treated surgically 93% of the time, while those with an acute Jones fracture were treated surgically only 62.7% of the time.

When Jones fractures are treated surgically, current literature supports intramedullary fixation involving a single screw, with or without biologic augmentation [9]. Of the patients treated operatively for a stress fracture in this systematic review, fixation was achieved using a single screw in 94.2% of patients and a plantar plate in 5.8% of patients. Biologics used to accelerate surgical healing included the use of a bone graft in 64.5% of patients and BMAC in 33.2% of patients included in this study. Unanimous consensus was reached in favour of the use of bone grafting during surgical fixation of fractures with nonunion or history of failed conservative treatment.

Acute Jones fractures may be more amenable to initial conservative treatment, even in elite athletes. Current literature favours conservative treatment of acute, nondisplaced Jones fractures with non-weight bearing cast immobilisation for 6–8 weeks [17]. The results of this systematic review found that athletes with acute fractures were more likely to be treated conservatively compared to those with stress etiologies. In this study, 62.7% of patients with an acute Jones fracture were treated surgically, compared to 37.3% who managed conservatively. Although the literature supports cast immobilisation as the gold-standard for conservative treatment [17], only 48.1% of patients treated conservatively in this study underwent non-weight bearing cast immobilisation, with the remaining patients treated with a walking boot (34.6%) or shock wave therapy (17.3%). Among this consensus group, there was unanimous agreement that immobilisation should occur in a walking boot rather than a stiff-soled shoe. Strong consensus was reached that athletes should remain non-weight bearing for at least 4–6 weeks, with return to partial weightbearing starting at 6 weeks.

A systematic review published by Attia et al. in 2021 demonstrated that athletes who were treated with intramedullary screw fixation for Jones fracture had a higher return to play rate and shorter time to return to play compared to those who were treated conservatively [2]. This systematic review found that a greater percentage of patients who were treated operatively achieved full union compared to those who were treated conservatively; in the acute fracture group, 96.5% of patients treated surgically achieved full union (165 out of 171) compared to 86.1% of patients who were treated conservatively (81 out of 94). In the stress fracture group, 90.1% of patients treated surgically achieved full union (572 out of 630) compared to 87.8% of patients treated conservatively (43 out of 49). Furthermore, the time to union and length of immobilisation was shorter among patients treated operatively compared to those who were treated nonoperatively. While all patients in the Jones stress fracture group returned to sport regardless of treatment intervention, one patient in the acute fracture group was unable to return to sport following surgical fixation [36].

Across both the acute and stress fracture groups, the average time to return to sport was shorter among athletes who were treated surgically at 11.9 ± 4.1 weeks compared to those who were treated conservatively at 14.6 ± 7.9 weeks. The results of this consensus process further support the findings of this systematic review; strong consensus was reached that athletes who undergo conservative treatment should expect to return to sport within 4–6 months, while those who undergo surgical treatment can expect to return to sport slightly sooner at 3–6 months.

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Refracture was the most common complication among patients treated surgically in the Jones stress fracture group. A total of six patients with stress fractures failed conservative treatment and underwent subsequent surgical fixation. In the acute Jones fracture group, hardware breakage and refracture were the most common complications reported among patients treated surgically, while chronic pain was the most common complication reported among patients managed conservatively. Four patients with acute fractures group failed conservative treatment and underwent subsequent surgical fixation. Hardware removal was the most common secondary surgical procedure in both the stress and acute Jones fracture groups.

This study was not without limitations. Although the results of the systematic review functioned to bolster or undermine each statement, some statements developed during this consensus could not be substantiated due to a lack of clinical data in the field. Efforts to increase the objectivity of participant selection included the use of standardised criteria to evaluate level of interest, degree of expertise, and academic reputation. While this consensus could not be completely void of bias, this initiative provides clinicians with evidence-based guidelines to aid in their management of a complex pathology.

## CONCLUSION

Among athletes with Jones fractures, the average time to return to sport was shorter among athletes who were treated surgically compared to those who were treated nonoperatively, regardless of injury aetiology or chronicity. This consensus process reached unanimous agreement with respect to the use of MRI for non-displaced fractures and CT for comminuted or displaced fractures, indications for intraoperative reduction and minimally invasive techniques, as well as recommendations for immobilisation and use of adjunctive bone grafting.

### AUTHOR CONTRIBUTIONS

**Scott D. Semelsberger**: Methodology; formal analysis and investigation; writing—original draft preparation. **Vanessa J. Boggiano**: Writing—original draft preparation. **Kassidy Webber**: Writing—original draft preparation. **Jay Moran**: Writing—review and editing. **Paul Mast**: Formal analysis and investigation. **Alberto Vasquez**: Formal analysis and investigation. **Lekha Yaramada**: Formal analysis and investigation; formal analysis and investigation. **Arianna L. Gianakos**: Conceptualisation; methodology; formal analysis and investigation; writing—review and editing; supervision. **IFASC Committee**: Formal analysis and investigation; writing—review and editing.

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### CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

### ETHICS STATEMENT

No Institutional Review Board or Ethics Committee approval was required due to the retrospective collection of publicly available data. All expert recommendations provided here are intended for interprofessional guidance and should not be interpreted as implying any liability.

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